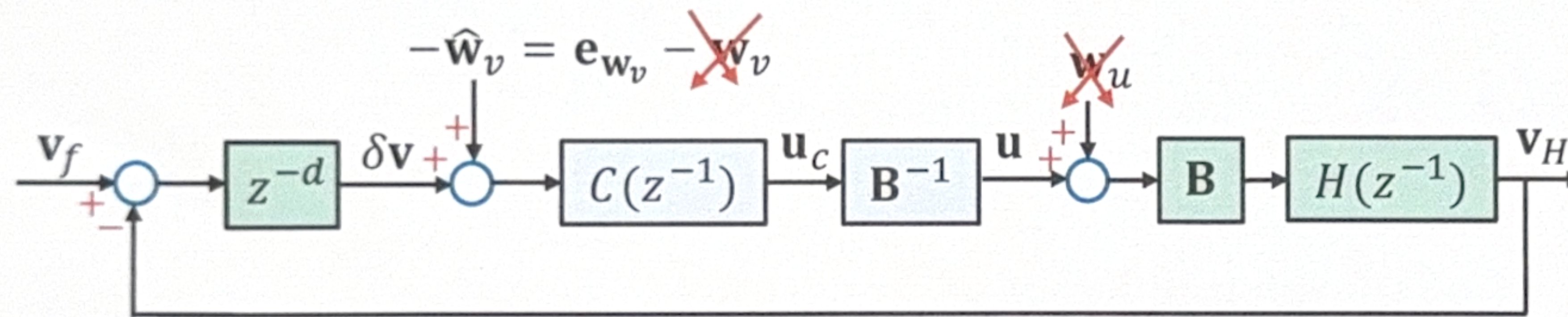
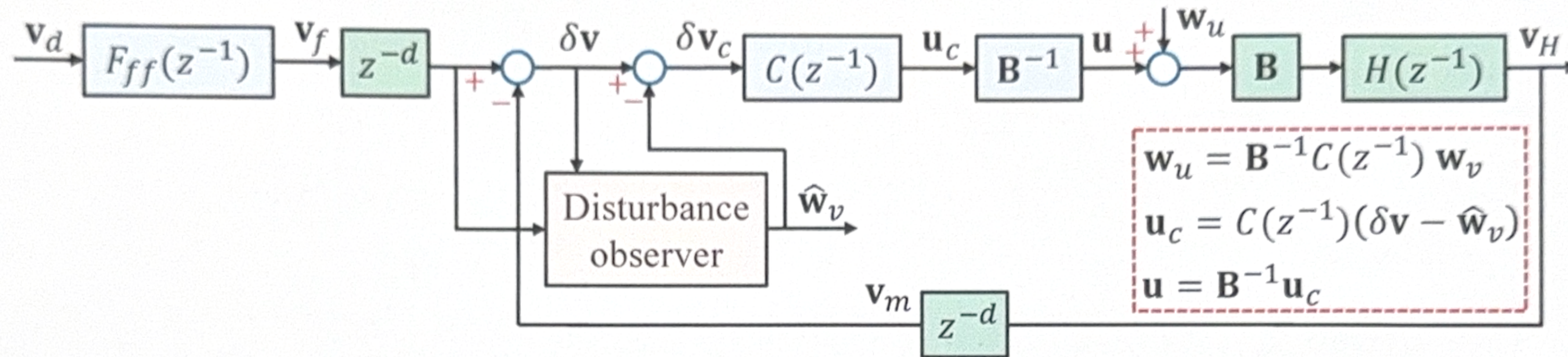
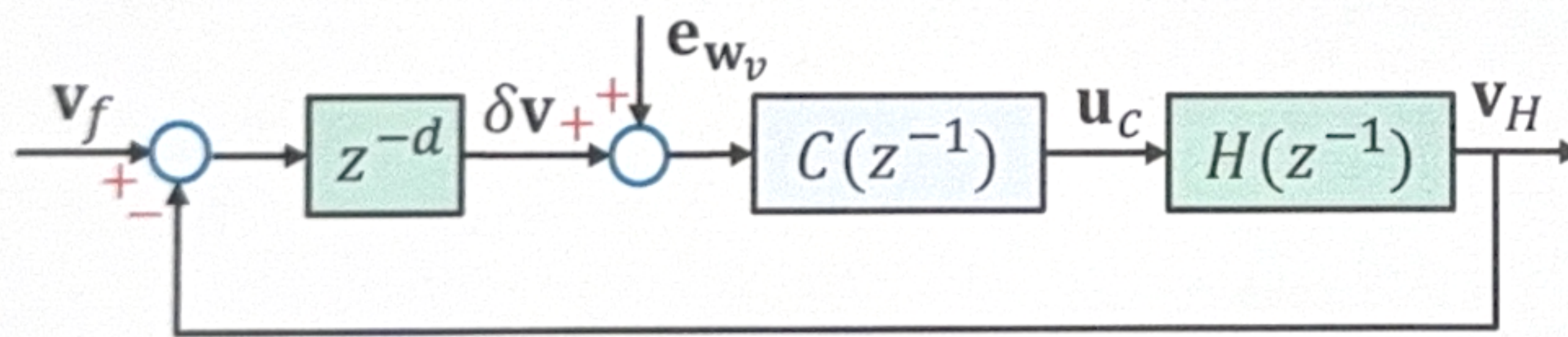
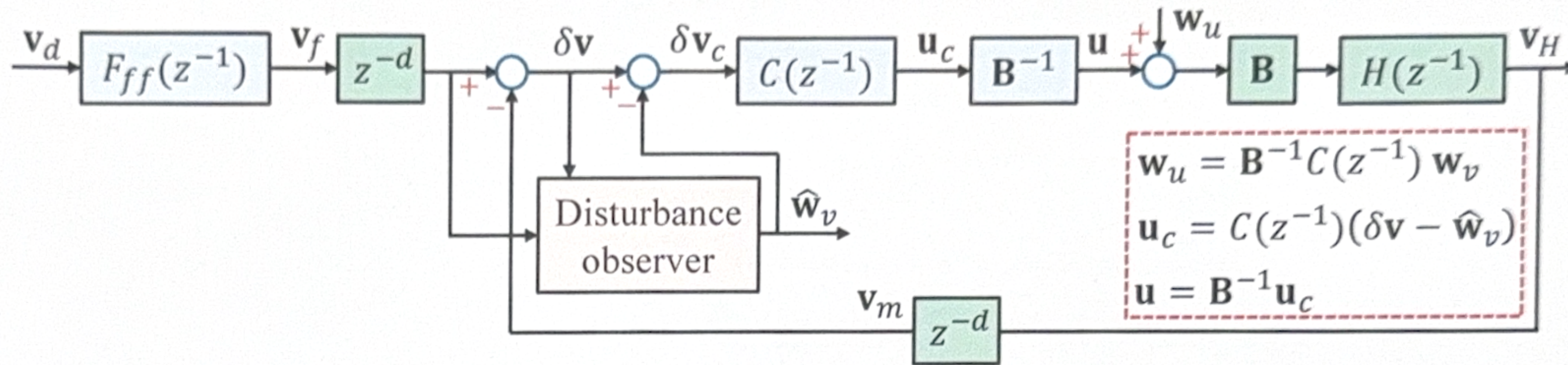




$$H(s) = \frac{1.1592 \times 10^7 B}{s^2 + 6.6172 \times 10^3 s + 1.1592 \times 10^7}$$

$$H(z^{-1}) = z^{-1} \frac{5.6695 \times 10^{-4} \times (1 + 0.9782z^{-1})B}{1 - 1.934848z^{-1} + 0.935970z^{-2}}$$





$$\frac{v_m}{u_c} = z^{-d} H(z^{-1}) = z^{-(1+d)} \frac{k_o(1+bz^{-1})}{(1-p_1 z^{-1})(1-p_2 z^{-1})} =$$

$$z^{-(1+d)} \frac{k_o(1+bz^{-1})}{1-a_1 z^{-1}-a_2 z^{-2}} = \frac{n_H(z^{-1})}{d_H(z^{-1})};$$

$$H_{cl}(z^{-1}) = \frac{z^{-d} v_H}{z^{-d} v_f} = \frac{v_H}{v_f} = \frac{z^{-d} C(z^{-1}) H(z^{-1})}{1+z^{-d} C(z^{-1}) H(z^{-1})} =$$

$$z^{-(1+d)} \frac{k_c(1+bz^{-1})}{1-\lambda_c z^{-1}}; k_c = \frac{1-\lambda_c}{1+b}$$

$$1 - H_{cl}(z^{-1}) = \frac{\delta v}{z^{-d} v_f} = 1 - z^{-(1+d)} \frac{k_c(1+bz^{-1})}{1-\lambda_c z^{-1}} =$$

$$\frac{1-\lambda_c z^{-1}-k_c z^{-(1+d)}-b_c z^{-(2+d)}}{1-\lambda_c z^{-1}}; b_c = b k_c = \frac{1-\lambda_c}{1+1/b}$$

$$C(z^{-1}) = \frac{1}{z^{-d} H(z^{-1})} \frac{H_{cl}(z^{-1})}{1-H_{cl}(z^{-1})} =$$

$$k_u \frac{1-a_1 z^{-1}-a_2 z^{-2}}{1-\lambda_c z^{-1}-k_c z^{-(1+d)}-b_c z^{-(2+d)}}; k_u = \frac{k_c}{k_o}$$

$$F_{ff}(z^{-1}) = k_{ff} \frac{1-\lambda_c z^{-1}}{1-\lambda_f z^{-1}}; k_{ff} = \frac{1-\lambda_f}{1-\lambda_c}$$

$$\textcircled{1} F_{ff}(z^{-1}) H_{cl}(z^{-1}) = \frac{v_H}{v_d} = k_f z^{-(1+d)} \frac{1+bz^{-1}}{1-\lambda_f z^{-1}}; k_f = \frac{1-\lambda_f}{1+b}$$

$$\frac{v_H}{v_d}(e^{-j\theta}) = A(\theta; \lambda_f, b) \angle \phi(\theta; \lambda_f, b) \quad (0 \leq \theta = \omega T \leq \pi);$$

$$A(\theta; \lambda_f, b) = k_f \left(\frac{1+2b \cdot \cos \theta + b^2}{1-2\lambda_f \cdot \cos \theta + \lambda_f^2} \right)^{1/2};$$

$$\phi(\theta; \lambda_f, b) = - \left[(1+d)\theta + \text{actan} \left(\frac{b \sin \theta}{1+b \cdot \cos \theta} \right) + \text{actan} \left(\frac{\lambda_f \sin \theta}{1-\lambda_f \cdot \cos \theta} \right) \right];$$

$$\textcircled{2} F_{ff}(z^{-1}) = k_{ff} \frac{1-\lambda_c z^{-1}}{1-\lambda_f z^{-1}} \frac{z^{-1}(1+bz)}{1} = k_{ff} \frac{b+(1-b\lambda_c)z^{-1}-\lambda_c z^{-2}}{1-\lambda_f z^{-1}}; k_{ff} = \frac{1-\lambda_f}{(1+b)(1-\lambda_c)};$$

$$F_{ff}(z^{-1}) H_{cl}(z^{-1}) = k_f z^{-(2+d)} \frac{1+b(z+z^{-1})+b^2}{1-\lambda_f z^{-1}}; k_f = \frac{1-\lambda_f}{(1+b)^2}$$

$$A(\theta; \lambda_f, b) = k_f \frac{1+2b \cdot \cos \theta + b^2}{\sqrt{1-2\lambda_f \cdot \cos \theta + \lambda_f^2}} \text{ and } \phi(\theta; \lambda_f) = - \left[(2+d)\theta + \text{actan} \left(\frac{\lambda_f \sin \theta}{1-\lambda_f \cdot \cos \theta} \right) \right]$$

$$A(\theta; 0, b) = k_f (1+2b \cdot \cos \theta + b^2) \text{ and } \phi(\theta; 0) = -(2+d)\theta$$

Design of Control Law

$$\mathbf{v}_f[k] = \begin{cases} \lambda_f \mathbf{v}_f[k-1] + k_{ff} \{\mathbf{v}_d[k] - \lambda_c \mathbf{v}_d[k-1]\}; & k_{ff} = \frac{1-\lambda_f}{1-\lambda_c} \\ \lambda_f \mathbf{v}_f[k-1] + k_{ff} \{b \mathbf{v}_d[k] + (1-b\lambda_c) \mathbf{v}_d[k-1] - \lambda_c \mathbf{v}_d[k-2]\}; & k_{ff} = \frac{1-\lambda_f}{(1+b)(1-\lambda_c)} \end{cases}$$

$$\delta \mathbf{v}_f[k] = \mathbf{v}_f[k-d] - \lambda_c \mathbf{v}_f[k-d-1] - k_c \mathbf{v}_f[k-2d-1] - b_c \mathbf{v}_f[k-2d-2]; k_c = \frac{1-\lambda_c}{1+b}; b_c = b k_c = \frac{1-\lambda_c}{1+1/b}$$

$$\delta \mathbf{v}[k] = \lambda_c \delta \mathbf{v}[k-1] + \delta \mathbf{v}_f[k] - k_c \{\mathbf{e}_{w_v}[k-d-1] + b \mathbf{e}_{w_v}[k-d-2]\}$$

Estimator ($d = 0$):

$$\delta \mathbf{v}_f[k] = \mathbf{v}_f[k] - \lambda_c \mathbf{v}_f[k-1] - k_c \mathbf{v}_f[k-1] - b_c \mathbf{v}_f[k-2]$$

$$\delta \hat{\mathbf{v}}[k] = \lambda_c \delta \hat{\mathbf{v}}[k-1] + \delta \mathbf{v}_f[k] + \ell_1 \mathbf{e}_{\delta v}[k-1]$$

$$\hat{\mathbf{w}}_1[k] = (1+\beta) \hat{\mathbf{w}}_1[k-1] - \beta \hat{\mathbf{w}}_2[k-1] + \ell_2 \mathbf{e}_{\delta v}[k-1]$$

$$\hat{\mathbf{w}}_2[k] = \hat{\mathbf{w}}_1[k-1] + \ell_3 \mathbf{e}_{\delta v}[k-1]$$

Uncoupled error dynamics (estimation):

$$\begin{bmatrix} e_{\delta v}[k] \\ e_{w1}[k] \\ e_{w2}[k] \end{bmatrix} = \begin{bmatrix} \lambda_c - \ell_1 & -k_c & -b k_c \\ -\ell_2 & 1 + \beta & -\beta \\ -\ell_3 & 1 & 0 \end{bmatrix} \begin{bmatrix} e_{\delta v}[k-1] \\ e_{w1}[k-1] \\ e_{w2}[k-1] \end{bmatrix}$$

$$\begin{cases} \ell_1 = \lambda_c + (1+\beta) - 3\lambda_e \\ \ell_2 = \frac{b(\lambda_e - 1)^3 - \beta(b+1)(\beta^2 - 3\beta\lambda_e + \beta + 3\lambda_e^2 - 3\lambda_e + 1)}{k_c(b+1)(b+\beta)} \\ \ell_3 = \frac{-\beta - b - \beta b + 3\beta\lambda_e + 3b\lambda_e - \beta^2 b - 3\lambda_e^2 - \beta^2 - \lambda_e^3 + 3\beta b \lambda_e}{k_c(b+1)(b+\beta)} \end{cases}$$

Estimator ($d = 1$):

$$\delta \mathbf{v}_f[k] = \mathbf{v}_f[k-1] - \lambda_c \mathbf{v}_f[k-2] - k_c \mathbf{v}_f[k-3] - b_c \mathbf{v}_f[k-4]$$

$$\delta \hat{\mathbf{v}}[k] = \lambda_c \delta \hat{\mathbf{v}}[k-1] + \delta \mathbf{v}_f[k] + \ell_1 \mathbf{e}_{\delta v}[k-1]$$

$$\hat{\mathbf{w}}_1[k] = (1+\beta) \hat{\mathbf{w}}_1[k-1] - \beta \hat{\mathbf{w}}_2[k-1] + \ell_2 \mathbf{e}_{\delta v}[k-1]$$

$$\hat{\mathbf{w}}_2[k] = \hat{\mathbf{w}}_1[k-1] + \ell_3 \mathbf{e}_{\delta v}[k-1]$$

$$\hat{\mathbf{w}}_3[k] = \hat{\mathbf{w}}_2[k-1] + \ell_4 \mathbf{e}_{\delta v}[k-1]$$

Uncoupled error dynamics (estimation):

$$\begin{bmatrix} e_{\delta v}[k] \\ e_{w1}[k] \\ e_{w2}[k] \\ e_{w3}[k] \end{bmatrix} = \begin{bmatrix} \lambda_c - \ell_1 & 0 & -k_c & -b k_c \\ -\ell_2 & 1 + \beta & -\beta & 0 \\ -\ell_3 & 1 & 0 & 0 \\ -\ell_4 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} e_{\delta v}[k-1] \\ e_{w1}[k-1] \\ e_{w2}[k-1] \\ e_{w3}[k-1] \end{bmatrix}$$

Implementation of Control Law ($d = 0$)

Input: $\mathbf{v}_d[k]$

Measurement: $\mathbf{v}_m[k]$ Acquired from ADC

Real time computation:

- ✓ 1. $\mathbf{v}_f[k] = \lambda_f \mathbf{v}_f[k-1] + k_{ff} \{ b \mathbf{v}_d[k] + (1 - b \lambda_c) \mathbf{v}_d[k-1] - \lambda_c \mathbf{v}_d[k-2] \}$
- ✓ 2. $\delta \mathbf{v}_f[k] = \mathbf{v}_f[k] - \lambda_c \mathbf{v}_f[k-1] - k_c \mathbf{v}_f[k-1] - b_c \mathbf{v}_f[k-2]$
3. $\delta \hat{\mathbf{v}}[k] = \lambda_c \delta \hat{\mathbf{v}}[k-1] + \delta \mathbf{v}_f[k] + \ell_1 \mathbf{e}_{\delta v}[k-1]$
4. $\hat{\mathbf{w}}_1[k] = (1 + \beta) \hat{\mathbf{w}}_1[k-1] - \beta \hat{\mathbf{w}}_2[k-1] + \ell_2 \mathbf{e}_{\delta v}[k-1]$
5. $\hat{\mathbf{w}}_2[k] = \hat{\mathbf{w}}_1[k-1] + \ell_3 \mathbf{e}_{\delta v}[k-1]$
6. $\delta \mathbf{v}[k] = \mathbf{v}_f[k] - \mathbf{v}_m[k]$
7. $\delta \mathbf{v}_c[k] = \delta \hat{\mathbf{v}}[k] - \hat{\mathbf{w}}_1[k]$
8. $\mathbf{u}_c[k] = \lambda_c \mathbf{u}_c[k-1] + k_c \mathbf{u}_c[k-1] + b_c \mathbf{u}_c[k-2] + k_u \{ \delta \mathbf{v}_c[k] - a_1 \delta \mathbf{v}_c[k-1] - a_2 \delta \mathbf{v}_c[k-2] \}$
9. $\mathbf{u}[k] = \mathbf{B}^{-1} \mathbf{u}_c[k]$ Sent to DAC
10. $\mathbf{e}_{\delta v}[k-1] = \delta \mathbf{v}[k] - \delta \hat{\mathbf{v}}[k]$
11. $\mathbf{v}_d[k-2] = \mathbf{v}_d[k-1]$
12. $\mathbf{v}_d[k-1] = \mathbf{v}_d[k]$
13. $\mathbf{v}_f[k-2] = \mathbf{v}_f[k-1]$
14. $\mathbf{v}_f[k-1] = \mathbf{v}_f[k]$
15. $\delta \hat{\mathbf{v}}[k-1] = \delta \hat{\mathbf{v}}[k]$
16. $\hat{\mathbf{w}}_1[k-1] = \hat{\mathbf{w}}_1[k]$
17. $\hat{\mathbf{w}}_2[k-1] = \hat{\mathbf{w}}_2[k]$
18. $\mathbf{u}_c[k-2] = \mathbf{u}_c[k-1]$
19. $\mathbf{u}_c[k-1] = \mathbf{u}_c[k]$
20. $\delta \mathbf{v}_c[k-2] = \delta \mathbf{v}_c[k-1]$
21. $\delta \mathbf{v}_c[k-1] = \delta \mathbf{v}_c[k]$

Implementation of Control Law ($d = 1$)

Input: $\mathbf{v}_d[k]$

Measurement: $\mathbf{v}_m[k]$ Acquired from ADC

Real time computation:

- ✓ 1. $\mathbf{v}_f[k] = \lambda_f \mathbf{v}_f[k-1] + k_{ff} \{ b \mathbf{v}_d[k] + (1 - b \lambda_c) \mathbf{v}_d[k-1] - \lambda_c \mathbf{v}_d[k-2] \}$
- ✓ 2. $\delta \mathbf{v}_f[k] = \mathbf{v}_f[k-1] - \lambda_c \mathbf{v}_f[k-2] - k_c \mathbf{v}_f[k-3] - b_c \mathbf{v}_f[k-4]$
3. $\delta \hat{\mathbf{v}}[k] = \lambda_c \delta \hat{\mathbf{v}}[k-1] + \delta \mathbf{v}_f[k] + \ell_1 \mathbf{e}_{\delta v}[k-1]$
4. $\hat{\mathbf{w}}_1[k] = (1 + \beta) \hat{\mathbf{w}}_1[k-1] - \beta \hat{\mathbf{w}}_2[k-1] + \ell_2 \mathbf{e}_{\delta v}[k-1]$
5. $\hat{\mathbf{w}}_2[k] = \hat{\mathbf{w}}_1[k-1] + \ell_3 \mathbf{e}_{\delta v}[k-1]$
6. $\hat{\mathbf{w}}_3[k] = \hat{\mathbf{w}}_2[k-1] + \ell_4 \mathbf{e}_{\delta v}[k-1]$
7. $\delta \mathbf{v}[k] = \mathbf{v}_f[k-1] - \mathbf{v}_m[k]$
8. $\delta \mathbf{v}_c[k] = \delta \hat{\mathbf{v}}[k] - \hat{\mathbf{w}}_1[k]$
9. $\mathbf{u}_c[k] = \lambda_c \mathbf{u}_c[k-1] + k_c \mathbf{u}_c[k-2] + b_c \mathbf{u}_c[k-3] + k_u \{ \delta \mathbf{v}_c[k] - a_1 \delta \mathbf{v}_c[k-1] - a_2 \delta \mathbf{v}_c[k-2] \}$
10. $\mathbf{u}[k] = \mathbf{B}^{-1} \mathbf{u}_c[k]$ Sent to DAC

12. $\mathbf{e}_{\delta v}[k-1] = \delta \mathbf{v}[k] - \delta \hat{\mathbf{v}}[k]$
13. $\mathbf{v}_d[k-2] = \mathbf{v}_d[k-1]$
14. $\mathbf{v}_d[k-1] = \mathbf{v}_d[k]$
15. $\mathbf{v}_f[k-3] = \mathbf{v}_f[k-2]$
16. $\mathbf{v}_f[k-2] = \mathbf{v}_f[k-1]$
17. $\mathbf{v}_f[k-1] = \mathbf{v}_f[k]$
18. $\delta \hat{\mathbf{v}}[k-1] = \delta \hat{\mathbf{v}}[k]$
19. $\hat{\mathbf{w}}_1[k-1] = \hat{\mathbf{w}}_1[k]$
20. $\hat{\mathbf{w}}_2[k-1] = \hat{\mathbf{w}}_2[k]$
21. $\hat{\mathbf{w}}_3[k-1] = \hat{\mathbf{w}}_3[k]$
22. $\mathbf{u}_c[k-3] = \mathbf{u}_c[k-2]$
23. $\mathbf{u}_c[k-2] = \mathbf{u}_c[k-1]$
24. $\mathbf{u}_c[k-1] = \mathbf{u}_c[k]$
25. $\delta \mathbf{v}_c[k-2] = \delta \mathbf{v}_c[k-1]$
26. $\delta \mathbf{v}_c[k-1] = \delta \mathbf{v}_c[k]$